

Gianmarco Santoro

Computer Vision - Machine Learning Engineer

Focused on Computer Vision, Robotics, Generative AI, LMM/LAM, and related areas

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DeepFake Detection | Computer Vision - Machine Learning Engineer

Media Integration and Communication Center (MICC) | Florence, Italy | 2023 - 2024

Implemented **Out-of-Distribution (OOD) Detection** on *FaceForensics++* dataset. A 3-step method:

- Extracted Self-Attention heatmaps via **Vision Transformer**, pre-trained on **face recognition**.
- Exploited heatmaps: trained a **Convolutional Autoencoder** on real ones.
- OODD: tested real vs fake reconstruction error, achieved **60% AUROC across 5 forgeries**, showing transfer learning capabilities.

Data Science & Statistical Learning | 2nd Level Master - Postgraduate

UniFi & IMT for Advanced Studies | 2023 - 2024 | 110/110 cum laude

Some topics: **Optimization**, GIS, NLP, Graph Theory. Projects (visit [Website](#) for more):

- Built **ML** housing price pipeline: K-Fold Cross-Validation, **Grid Search** on polynomial regression degree improving MAE by 12%.
- Developed **Deep Learning images classifier** via **CNNs** models (*PyTorch*), achieving an accuracy of 70.4% on the test set.
- **Predicted salary** on selected features using **CART** and other models. Reached 83.8% accuracy with Gradient Boosting.
- Forecasted post-booking classes attendance: achieved 75.8% accuracy with **logistic regression** on a highly imbalanced dataset.

Smart Center | IT & System Integration Engineer

Daikin Applied Europe SpA | Rome, Italy | 2021 - 2022

Worked on cloud-based HVAC monitoring, system integration, and services:

- **Automated PLC** mapping and alarm selection via **Python**, reducing manual configuration time by 95%.
- Built real-time IoT sensor-to-cloud **temperature control system** integrating 21 Modbus sensors, removing 100% manual setup.
- Aggregated data via web API to generate 2000+ reports/month, improving issues detection and reducing workload by 80%.

Wireless Charging System | Operations Research Scientist

Enermove Srl | Turin, Italy | 2020 - 2021

Optimized electric vehicles wireless charging system placement via **Genetic Algorithms**, improving energy efficiency-to-cost ratio.

Chip Extrusion Recycling | Manufacturing Research Engineer

Institut für Umformtechnik und Leichtbau | Dortmund, Germany | 2018

Experimented with the **aluminum recycling process** by direct extrusion of mixed waste alloys (DoE). Compared properties to simulated ones (FEM and numerical). Assessed **industrial feasibility** to identify process bottlenecks, such as logistic ones and outcomes defects.

Mechanical Engineering | M.Sc. at PoliTo & B.Sc. at UniBo | 104/110

Specialized in **Automation**. Some extra projects: • Automation and **mechatronics** labs: PLC, PID controls, pneumatics, mechanisms.

- Designed a reduction **gearbox**. • Developed a home **heating system**. • Built a **numerical model** for metal flue heat exchange.

Personal Projects | 2022 - Current

- Developed my **Website**. Tools: *HTML, CSS, JS, VScode, GIT & GitHub*, tested across *Arc, Safari, and Chrome* browsers.
- Conducted Market Research, Javelin Board and SWOT analysis, Lean Startup Method on MVPs: **Project-Driven** Work Community, **Food-based** Social Media, **Selected Products** Referral Platform. Tools: *Figma, Mural, Notion, Canva, G-Forms/Sheets/Docs*.

Skills | Languages: • **Italian:** C2 - Mother tongue • **English:** C1 - Professional • **German:** A2 - Basic

AI & SW: Python, PyTorch, OpenCV, NumPy, Pandas, Scikit-learn, Matplotlib, Seaborn, GIT-GitHub/Lab, R, Matlab & Simulink

Web, Tools, Industrial: HTML, CSS, Jupyter/Colab, MS-Office (Excel), G-Suite, Notion, Canva, AutoCAD, SolidWorks

Basic: Bash, C, JS, PLC Siemens-Clim., TCP/IP, SSH, mRemoteNG, Modbus, qModMaster, CATIA, HyperMesh/Ext., MSC-Adams/Apex